

Notes: Eizenberg (RES, 2014)

Upstream Innovation and Product Variety in the U.S. Home PC Market

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1 Big picture

- **Question.** Does rapid upstream innovation increase welfare for everyone, or can it crowd out older downstream products and leave price-sensitive consumers worse off in the short run?
- **Market.** The paper studies the U.S. home personal computer market from 2001Q3 to 2004Q2, focusing on how CPU innovations from Intel affected the variety of PCs offered by downstream PC makers.
- **Central innovation.** The landmark upstream product is Intel’s Pentium M chip, important for mobile computing and introduced during the rise of notebooks.
- **Core economic mechanism.** An upstream innovation expands the feasible set for PC makers, but downstream firms face fixed costs of offering configurations. They may drop older/basic products when new products arrive.
- **Main empirical framework.** The paper estimates demand, marginal costs, and fixed-cost bounds in a two-stage game in which PC firms first choose product configuration portfolios and then set prices.
- **Key methodological challenges.** Product choices are multiple discrete choices, equilibrium may be non-unique, and fixed-cost shocks are observed by firms but not by the econometrician, creating sample-selection concerns.
- **Main findings.** The Pentium M increased total consumer surplus by about 3.2–6.3%, substantially increased mobile PC sales, and crowded out older Pentium III and Pentium 4 notebook configurations.
- **Distributional result.** The benefits of innovation are concentrated among the 20% least price-sensitive consumers. More price-sensitive consumers benefit little from the innovation in equilibrium because older affordable notebooks disappear.
- **Counterfactual product-variety result.** Adding older Pentium III-based notebooks back to the 2004Q2 market would not clearly raise total welfare because extra consumer surplus and variable profit are offset by fixed costs, but it would create affordable laptop options and redistribute innovation benefits to price-sensitive households.
- **Takeaway.** Upstream innovation can raise aggregate welfare while narrowing downstream product variety in ways that limit short-run access for price-sensitive consumers.

2 Incremental contribution of the paper

- **Endogenous product-variety contribution.** Earlier new-goods welfare papers often treated product availability as exogenous. This paper endogenizes both the number and identity of downstream product configurations offered by PC makers.
- **Innovation-versus-variety contribution.** The paper is not only about valuing a new product. It studies whether a frontier upstream innovation crowds out older products and thereby changes who benefits from innovation.
- **Multiple-equilibria and partial-identification contribution.** The paper handles a product-choice game in which equilibrium need not be unique and fixed costs are partially identified. It then uses potential equilibria in counterfactual simulations.
- **Selection-on-product-offerings contribution.** Fixed-cost shocks are observed by firms when they decide which configurations to offer but are not observed by the econometrician. The paper explicitly addresses the resulting selection issue instead of assuming it away.
- **Distributional-welfare contribution.** The paper computes welfare effects by consumer price-sensitivity groups and shows that aggregate innovation benefits can be highly skewed toward high-valuation consumers.
- **Platform/upstream-downstream contribution.** The paper connects upstream technological innovation by Intel to downstream firms’ endogenous portfolios, pricing, and consumer welfare in the PC market.

- **Policy contribution.** It shows why a policy that preserves older product varieties may have little aggregate welfare gain but large distributional effects by creating affordable product categories.

3 Data and measurement

3.1 Market and product data

The paper uses IDC data on quarterly PC sales in the U.S. home market from 2001Q3 through 2004Q2. Each observation is a PC product configuration, which combines a PC product line with a CPU technology. A configuration is defined by vendor, product line, CPU brand, CPU generation, CPU speed range, desktop/notebook status, and price.

- The main sample contains 2,287 quarter-product observations.
- Very small products with sales below 100 units per quarter are excluded for computational reasons.
- Apple products are excluded because the author lacks processor speed information for them.
- Desktops and notebooks are both included in price equilibrium estimation, though the main product-variety counterfactual focuses on leading notebook product lines.

Interpretation: The data are rich enough to observe both product variety and price, but product definitions are necessarily discrete. This is why the model treats CPU configurations as the object of downstream product-choice decisions.

3.2 Market size and outside option

The market size is based on potential home PC consumers. The outside option includes not buying a new PC in the current quarter. The demand model estimates how consumer utility from all inside products changes over time, allowing the value of a fixed bundle of characteristics to decline as overall PC technology improves.

Interpretation: The outside option is central in a fast-improving durable-goods market. A product that was attractive in 2001 may have much lower value in 2004 even if its nominal characteristics are unchanged.

3.3 Product characteristics

The main product characteristics include:

- price;
- desktop versus notebook indicator;
- CPU speed ranges;
- CPU brand and generation indicators, such as Celeron, Pentium III, Pentium 4, and Pentium M;
- PC vendor and product-line/brand indicators;
- a time trend capturing the declining value of a fixed bundle of characteristics.

Interpretation: The CPU variables are the bridge between upstream innovation and downstream product choice. The paper can ask how the introduction of a chip changes product portfolios because CPU technologies are observed in product-level sales data.

3.4 Demand-side heterogeneity

The paper estimates a random-coefficients logit model with heterogeneity in price sensitivity and in tastes for product attributes such as portability and CPU features. Price sensitivity is log-normal so that simulated consumers have the correct sign of price response.

Interpretation: Heterogeneity is not a side detail. It is what lets the paper distinguish the effects of innovation on high-valuation and low-valuation consumers.

3.5 Cost-side objects

Marginal costs are inferred from pricing first-order conditions in the second-stage pricing game. Fixed costs of offering configurations are not point-identified because firms' product choices depend on fixed-cost shocks observed by firms but unobserved by the econometrician. The paper therefore estimates bounds on firm-level mean fixed costs.

Interpretation: The separation between marginal costs and fixed costs is essential. Marginal costs determine pricing incentives conditional on products being offered; fixed costs determine whether products are offered at all.

4 Model and empirical specifications

4.1 Demand system

The indirect utility of household i from product j in quarter t can be represented as

$$u_{ijt} = x'_{jt}\beta_i - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt},$$

where x_{jt} includes product characteristics, p_{jt} is price, ξ_{jt} is an unobserved product-level demand shock, and ε_{ijt} is an idiosyncratic logit error.

Interpretation: The demand model generates own- and cross-price elasticities, willingness-to-pay measures, and variable profits under alternative product portfolios.

4.2 Two-stage game

The paper models PC makers as playing a two-stage game in each quarter.

1. **Stage 1: product configuration choice.** Firms observe fixed-cost shocks and choose which CPU configurations to offer with each product line.
2. **Stage 2: price setting.** After product choices are fixed, firms observe demand and marginal-cost shocks and set prices in a Nash-Bertrand pricing game.

Interpretation: The timing reflects the idea that product portfolio decisions are commitments made before all pricing shocks are realized.

4.3 Fixed costs

The fixed cost of configuration j for firm d is

$$F_j = F_d + \nu_j,$$

where F_d is a firm-level mean fixed cost and ν_j is a mean-zero fixed-cost shock observed by the firm before product choice.

Interpretation: Because ν_j is observed by the firm but not by the econometrician, conditioning on observed product choices creates selection. A product that is offered may have a favorable fixed-cost draw.

4.4 Marginal costs and pricing

Marginal cost depends on observed cost shifters and an unobserved marginal-cost shock:

$$\log(mc_j) = w'_j \gamma + \omega_j.$$

Given a product portfolio, firm d 's variable profit is

$$\pi_d = \sum_{m \in S_d} \sum_{\ell \in L_{dm}} [p_{m\ell} - mc_{m\ell}] s_{m\ell}(p) M - TF_d.$$

The Nash-Bertrand pricing first-order condition can be written as

$$p - mc = (T * \Delta(p; \theta_2))^{-1} s(p),$$

where T is the ownership matrix and Δ is the matrix of market-share derivatives.

Interpretation: This links demand estimates to marginal costs and variable profits. Variable profits are the object compared with fixed costs in the product-choice inequalities.

4.5 Product-choice inequalities and fixed-cost bounds

For an observed product j that firm d offers, removing j must not be profitable. Thus the fixed-cost saving from dropping j cannot exceed the loss in expected variable profit. For an unoffered product, adding it must not be profitable, so its fixed cost must exceed the expected variable-profit gain.

Interpretation: These necessary inequalities produce bounds on fixed costs. Because fixed-cost shocks create selection, the paper uses bounded-support assumptions to derive inequalities involving unconditional rather than conditional fixed-cost shocks.

4.6 Potential equilibria in counterfactuals

Since fixed costs are set-identified and product-choice games can have multiple equilibria, the counterfactual does not produce a single outcome. The paper computes a set of potential equilibria: product portfolios that cannot be ruled out as equilibrium outcomes for some fixed-cost values in the estimated set.

Interpretation: Counterfactual welfare effects are therefore reported as ranges. This is why many tables report lower and upper bounds rather than single estimates.

5 Identification strategy

5.1 Demand identification

Demand identification uses variation in prices, product characteristics, vendor/brand indicators, CPU technologies, and time. Prices are endogenous, so the paper compares logit OLS and logit IV estimates and then estimates a BLP random-coefficients model. Instruments follow the differentiated-products demand literature and use variation in product characteristics across rival products and market structure.

Interpretation: The logit IV price coefficient is much more negative than the OLS coefficient, showing the importance of correcting price endogeneity.

5.2 Marginal cost identification

Given demand estimates, the second-stage Bertrand first-order conditions identify marginal costs. The ownership matrix maps products to firms and determines how firms internalize substitution among their own products.

Interpretation: Marginal costs are not observed directly; they are backed out from prices and demand elasticities. Therefore, elasticities are crucial for both cost and welfare calculations.

5.3 Fixed-cost partial identification

Fixed costs are identified through necessary equilibrium inequalities. The key challenge is selection: firms choose products after observing fixed-cost shocks. Rather than assuming away selection, the paper imposes bounded support on fixed costs and obtains intervals for firm-level mean fixed costs.

Interpretation: This is a core methodological contribution. It avoids a false sense of precision in product-choice models with unobserved fixed costs and multiple equilibria.

5.4 Counterfactual identification

The main counterfactual removes Pentium M technology from the feasible set in 2004Q2 and computes potential equilibrium outcomes. The comparison of observed outcomes with no-Pentium-M potential equilibria identifies the effect of the upstream innovation, subject to the model and feasible-set assumptions.

Interpretation: The paper treats Intel’s technology menu as exogenous to downstream PC makers. This is plausible because Intel was the dominant platform leader and CPU innovation followed Moore’s Law-like technological progress.

5.5 Limitations

- The model is static and does not include consumer durability or forward-looking purchase timing.
- Desktop offerings are treated as exogenous in the main notebook counterfactual, though desktop prices remain endogenous.
- AMD-based notebook offerings are held fixed because some leading notebook producers did not use AMD chips, limiting identification of AMD fixed costs.

- The potential-equilibria set captures what cannot be ruled out rather than a single equilibrium selection rule.

Interpretation: These limitations matter because the PC market is durable and forward-looking. Still, the static model isolates the product-variety mechanism in a tractable way.

6 Figures and tables

6.1 Tables 1–2: Top vendors’ market shares and CPU vendor shares

Read: Table 1 shows that the top five PC vendors account for about 60–70% of the U.S. home PC market over the sample period. Table 2 shows that Intel supplies more than 70% of CPUs used in the home PC market.

Interpretation: The downstream PC market is moderately concentrated, while the upstream CPU market is highly concentrated. This institutional structure motivates the paper’s upstream/downstream framework.

Economic message: Intel’s technological changes can affect many downstream firms because Intel dominates the CPU input market.

TABLE 1
Top vendors’ market shares, U.S. Home PC market

Year 1		Year 2		Year 3	
Vendor	Share	Vendor	Share	Vendor	Share
Dell	0.190	Dell	0.263	Dell	0.279
HP ^a	0.185	HP	0.234	HP	0.258
Compaq ^a	0.092	eMachines	0.076	eMachines ^a	0.070
Gateway	0.091	Gateway	0.070	Gateway ^b	0.053
eMachines	0.060	Toshiba	0.042	Toshiba	0.043
Top 5 vendors	0.618	Top 5 vendors	0.685	Top 5 vendors	0.704

Note: Years: 01Q3-02Q2, 02Q3-03Q2, 03Q3-04Q2. ^aCompaq and HP merge in Year 1, eMachines and Gateway merge in Year 3.

(a) Table 1: Top vendors’ market shares, U.S. Home PC market.

TABLE 2
CPU vendor shares

Vendor	Market shares		
	Year 1	Year 2	Year 3
Intel	0.71843	0.72246	0.74496
AMD	0.24429	0.23643	0.22032
IBM	0.03230	0.03450	0.03048
Others	0.00477	0.00524	0.00323
Transmeta	0.00022	0.00135	0.00097
Via	0.00000	0.00002	0.00005

Note: Years: 01Q3-02Q2, 02Q3-03Q2, 03Q3-04Q2, U.S. Home market.

(b) Table 2: CPU vendor shares.

6.2 Figures 1–2: CPU speed range shares and average prices/portable shares

Read: Figure 1 shows rapid movement toward faster CPU speed ranges. Figure 2 shows declining average prices and rising portable shares over the sample period.

Interpretation: The figures document the pace of innovation and the shift toward mobility. They motivate the focus on Pentium M as a mobile computing innovation.

Economic message: The market is not static: product quality improves and prices fall, making it important to model both innovation and endogenous product replacement.

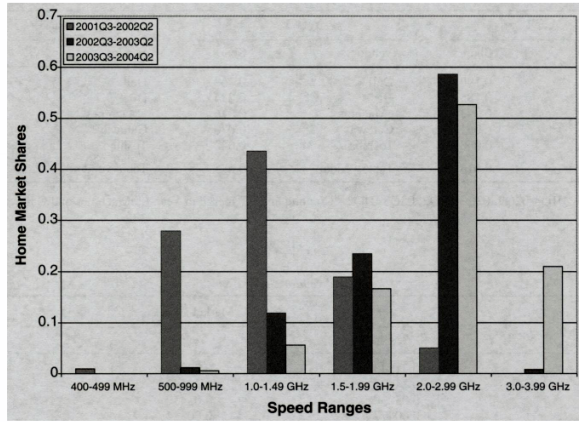


FIGURE 1
CPU speed range shares, U.S. Home Market, over the three sample years

(a) Figure 1: CPU speed range shares.

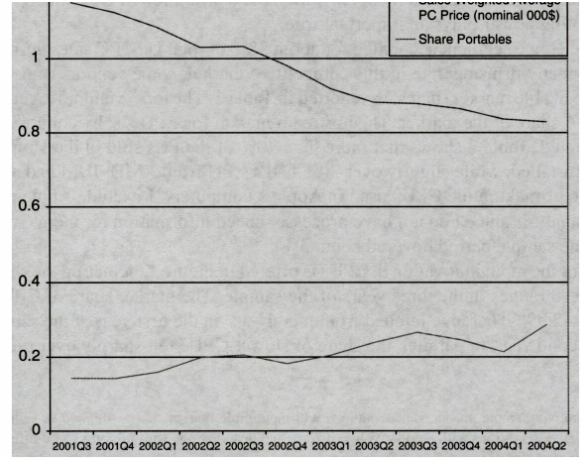


FIGURE 2
Sales-weighted average prices (\$1000's) and share portables, U.S. Home PC Market

(b) Figure 2: Prices and portable shares.

TABLE 3
Adoption rates of CPU technologies

(a) Desktop product lines					
Quarter	C_0.5-0.99	C_1-1.49	C_1.5-1.99	C_2-2.99	P3_0.5-0.99
2001Q3	0.89	0.00	0.00	0.00	0.93
2001Q4	0.46	0.42	0.00	0.00	0.46
2002Q1	0.35	0.58	0.00	0.00	0.31
2002Q2	0.13	0.57	0.00	0.00	0.17
2002Q3	0.09	0.39	0.48	0.13	0.13
2002Q4	0.07	0.04	0.44	0.41	0.11
2003Q1	0.04	0.04	0.41	0.41	0.04
2003Q2	0.04	0.04	0.37	0.41	0.04
2003Q3	0.04	0.04	0.24	0.48	0.04
2003Q4	0.04	0.04	0.20	0.52	0.04
2004Q1	0.00	0.04	0.15	0.54	0.00
2004Q2	0.00	0.00	0.17	0.54	0.00
Quarter	P3_1-1.49	P4_1-1.49	P4_1.5-1.99	P4_2-2.99	P4_3-3.99
2001Q3	0.67	0.48	0.26	0.00	0.00
2001Q4	0.50	0.65	0.65	0.12	0.00
2002Q1	0.31	0.58	0.73	0.50	0.00
2002Q2	0.13	0.43	0.70	0.65	0.00
2002Q3	0.13	0.26	0.74	0.70	0.00
2002Q4	0.04	0.11	0.37	0.81	0.00
2003Q1	0.00	0.11	0.44	0.81	0.15
2003Q2	0.00	0.11	0.44	0.81	0.15
2003Q3	0.00	0.08	0.28	0.92	0.60
2003Q4	0.00	0.12	0.28	0.92	0.60
2004Q1	0.00	0.08	0.19	0.92	0.65
2004Q2	0.00	0.08	0.17	0.92	0.63

(b) Portable product lines						
Quarter	C_0.5-0.99	C_1-1.49	C_1.5-1.99	C_2-2.99	P3_0.5-0.99	P3_1-1.49
2001Q3	0.81	0.00	0.00	0.00	1.00	0.15
2001Q4	0.59	0.21	0.00	0.00	0.79	0.72
2002Q1	0.36	0.25	0.00	0.00	0.64	0.86
2002Q2	0.12	0.31	0.00	0.00	0.54	0.62
2002Q3	0.11	0.21	0.07	0.00	0.18	0.64
2002Q4	0.10	0.03	0.23	0.10	0.16	0.42
2003Q1	0.03	0.06	0.26	0.13	0.19	0.39
2003Q2	0.03	0.03	0.21	0.12	0.15	0.42
2003Q3	0.03	0.00	0.25	0.13	0.16	0.34
2003Q4	0.03	0.03	0.22	0.13	0.13	0.28
2004Q1	0.00	0.03	0.18	0.18	0.03	0.18
2004Q2	0.00	0.03	0.19	0.19	0.03	0.19
Quarter	P4_1-1.49	P4_1.5-1.99	P4_2-2.99	P4_3-3.99	Pm_1-1.49	Pm_1.5-1.99
2001Q3	0.00	0.00	0.00	0.00	0.00	0.00
2001Q4	0.00	0.00	0.00	0.00	0.00	0.00
2002Q1	0.07	0.18	0.00	0.00	0.00	0.00
2002Q2	0.19	0.38	0.00	0.00	0.00	0.00
2002Q3	0.14	0.46	0.32	0.00	0.00	0.00
2002Q4	0.10	0.52	0.58	0.00	0.00	0.00
2003Q1	0.13	0.52	0.58	0.00	0.10	0.06
2003Q2	0.12	0.48	0.55	0.00	0.09	0.09
2003Q3	0.09	0.53	0.59	0.06	0.22	0.19
2003Q4	0.13	0.50	0.50	0.09	0.31	0.25
2004Q1	0.12	0.42	0.52	0.12	0.21	0.48
2004Q2	0.06	0.44	0.50	0.13	0.28	0.56

Notes: Home market, Intel technologies. Excluding vendors identified as "Others", Apple products, and products with sales smaller than 100 units in a quarter. C=Celeron, P3=Pentium III, P4=Pentium 4 and Pm=Pentium M. P3_0.5-0.99=Pentium III with speed between 0.5 and 0.99 GHz. As explained in the text, the "adoption rate" of a technology is different than its market share. To demonstrate this concept, consider the C_0.5-0.99 example: there were 28 desktop product lines in 2001Q3, 25 of which offered this CPU technology, implying an adoption rate of $25/28=0.89$. CPU technologies with very small installation rates are listed in panel (b).

Table 3: Adoption rates of CPU technologies.

TABLE 4
The sets H of feasible CPU technologies

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
P3_0.5-0.99	X	X	X									
P3_1.0-1.49	X	X	X	X	X	X	X	X	X	X		
C_0.5-0.99	X	X	X									
C_1.0-1.49		X	X	X	X							
C_1.5-1.99					X							
C_2-2.99					X	X	X	X	X	X	X	X
P4_1.0-1.49				X	X	X	X	X	X	X	X	X
P4_1.5-1.99			X	X	X	X	X	X	X	X	X	X
P4_2-2.99					X	X	X	X	X	X	X	X
P4_3-3.99						X	X	X	X	X	X	X
PM_1.0-1.49									X	X	X	X
PM_1.5-1.99									X	X	X	X

Notes: The table shows the sets of the 12 data quarters that set H of feasible CPU technologies. For each feasible technology, the table shows the quarters in which it was available.

Table 4: Feasible CPU technologies.

6.3 Table 3: Adoption rates of CPU technologies

Read: Table 3 reports adoption rates of CPU technologies by quarter for desktop and portable product lines.

Interpretation: Adoption varies substantially across CPU generations and quarters. Pentium M appears later and becomes important in portable lines.

Economic message: Upstream CPU technologies diffuse unevenly into downstream product portfolios, which is exactly what the product-choice model seeks to explain.

6.4 Table 4: The sets H of feasible CPU technologies

Read: Table 4 reports which CPU technologies are feasible in each quarter according to sample-based inclusion criteria.

Interpretation: The feasible set is central to counterfactual product-choice analysis. A firm can only choose from technologies that are commercially available and relevant in that quarter.

Economic message: Upstream innovation operates by changing the menu of feasible downstream product configurations.

6.5 Tables 5–6: Logit and BLP demand estimates

Read: Table 5 shows that IV demand estimates imply much stronger price sensitivity than OLS. Table 6 reports the main BLP random-coefficients estimates, including heterogeneity in price sensitivity and taste parameters.

Interpretation: The estimates show strong consumer heterogeneity. Consumers value CPU speed and CPU brand, while the negative time trend implies that the utility of a fixed bundle falls quickly as the market frontier improves.

Economic message: Heterogeneity is necessary to understand why innovation benefits high-valuation consumers but does not automatically benefit price-sensitive consumers.

β	Logit_OLS	Logit_IV
Price (00\$)	-0.0395*** (0.0135)	-0.157** (0.0649)
Laptop dummy	-0.616*** (0.0999)	-0.298 (0.199)
Trend	-0.0398** (0.0171)	-0.138** (0.0567)
CPU speed range dummies		
1-1.49 GHz	0.200* (0.107)	0.385** (0.152)
1.5-1.99 GHz	0.383*** (0.138)	0.660*** (0.208)
2-2.99 GHz	0.752*** (0.156)	1.223*** (0.303)
3-3.99 GHz	0.779*** (0.253)	1.586*** (0.508)
CPU brand dummies		
AMD Duron	0.694*** (0.208)	0.544** (0.254)
AMD Athlon	0.691*** (0.115)	0.695*** (0.133)
Intel Pentium III	0.227** (0.116)	0.507*** (0.189)
Intel Pentium 4	0.359*** (0.103)	0.629*** (0.176)
Intel Pentium M	0.724*** (0.215)	1.554*** (0.489)
Constant	-10.66*** (0.183)	-9.441*** (0.699)
Observations	2287	2287
R^2	0.491	0.473

Notes: Standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Dummy variables for main PC vendors and brands included, not reported.

(a) Table 5: Descriptive logit demand results.

(a) Main PC characteristics									
	β	SE	σ	SE	γ	SE			
Constant	4.479	3.108	1.546	1.933	6.759	0.020			
Laptop Dummy	-0.690	1.158	3.785	0.518	0.312	0.013			
Trend	-1.444	0.263	0.430	0.081	-0.090	0.002			
CPU speed range dummies									
1-1.49 GHz	2.390	0.386			0.156	0.013			
1.5-1.99 GHz	3.621	0.521			0.232	0.016			
2-2.99 GHz	6.212	0.809			0.412	0.017			
3-3.99 GHz	9.584	1.374			0.709	0.030			
CPU brand dummies									
AMD Duron	-0.915	0.443			-0.120	0.023			
AMD Athlon	0.912	0.217			0.031	0.013			
Intel Pentium III	3.517	0.484			0.272	0.014			
Intel Pentium 4	3.855	0.487			0.305	0.010			
Intel Pentium M	10.051	1.361			0.741	0.032			
Price sensitivity									
	α	SE	σ^p	SE					
	0.810	0.179	0.301	0.060					
(b) PC vendor and brand dummies									
	β	SE	γ	SE	β	SE	γ	SE	
Dell	12.332	2.603	0.774	0.062	eMachines	0.389	0.602	-0.325	0.050
dimension	-10.426	2.813	-0.915	0.065	Toshiba	7.933	1.684	0.479	0.050
inspire	-9.908	2.732	-0.838	0.064	portage	0.593	1.018	0.093	0.059
latitude	-7.529	2.137	-0.488	0.071	port_tablet	2.855	1.303	0.218	0.085
optiPlex	-13.509	2.819	-0.903	0.064	satellite	-5.405	1.752	-0.517	0.055
HP	-0.976	0.334	-0.049	0.021	satpro	-2.628	1.141	-0.132	0.053
evoipaq	-1.651	0.519	-0.174	0.030	Sony	5.684	0.821	0.306	0.037
media	7.568	0.870	0.424	0.029	vaio_ds	-3.909	0.871	-0.265	0.043
pavilion	2.625	0.385	-0.015	0.024	vaio_r	0.500	0.824	0.205	0.052
presario	2.593	0.355	0.026	0.020	vaio_w	3.163	0.979	0.283	0.059
cmpq_notebook	1.841	0.653	0.175	0.033	vaio_505	1.288	0.963	-0.007	0.061
cmpq_ultrabk	10.945	2.221	0.741	0.080	vaio_fx	0.951	0.734	0.053	0.053
Gateway	0.309	0.399	0.068	0.025	IBM	2.037	1.217	0.208	0.083
gateway3	-2.619	0.730	-0.408	0.035	netvista	-3.868	1.307	-0.244	0.087
gateway5	1.755	0.865	-0.030	0.048	thinkCentre	0.419	1.301	0.040	0.095
gateway7	2.159	0.690	0.077	0.035	thinkpadA	8.348	2.084	0.452	0.097
essential	2.124	0.458	-0.098	0.030	thinkpadT	1.253	1.366	-0.016	0.092
performance	1.751	0.530	0.039	0.034	thinkpadR	-3.304	1.291	-0.233	0.085
media	4.960	0.828	0.365	0.035	Acer_veriton	-2.120	0.382	-0.120	0.016
gateway4	-1.320	0.510	-0.139	0.031	Averatec	1.131	0.688	-0.034	0.048
gateway6	4.725	1.077	0.173	0.051	Fujitsu	-1.090	0.354	-0.018	0.023
solo	0.185	0.868	-0.106	0.050	MicroElectronics	-1.585	0.236	-0.009	0.017

Notes: Obs: 2287. Standard errors are not corrected for simulation error, which is mitigated via antithetic draws. See text

(b) Table 6: BLP estimates for demand and cost parameters.

6.6 Tables 7a-7b: Demand elasticities, markups, and willingness to pay

Read: Table 7a reports own- and cross-price elasticities for a sample of desktops and notebooks. Table 7b reports willingness to pay for CPU and brand upgrades, median markup, and correlations among markups, prices, and shocks.

Interpretation: Cross-price elasticities are stronger within notebook products than across desktops and notebooks. Table 7b shows large willingness to pay for the Pentium M upgrade and a negative value of moving one year forward with a fixed bundle.

Economic message: The estimates imply strong product differentiation and rapid obsolescence. The market values innovation, but old products depreciate quickly in consumers' utility.

	Dell	HP	Gateway	Sony	IBM	Dell ^a	Gateway ^a	HP ^a	IBM ^a	Toshiba ^a
Dell Dimension	-9.07	0.26	0.10	0.03	0.00	0.05	0.00	0.02	0.00	0.04
HP Pavilion	0.25	-10.43	0.14	0.05	0.01	0.05	0.00	0.02	0.00	0.04
Gateway 5 Series	0.17	0.25	-11.96	0.09	0.01	0.04	0.00	0.02	0.00	0.04
Sony VAIO R	0.10	0.20	0.20	-13.22	0.01	0.03	0.00	0.02	0.00	0.03
IBM ThinkCentre	0.17	0.25	0.17	0.09	-12.10	0.04	0.00	0.02	0.00	0.04
Dell Inspiron ^a	0.07	0.07	0.03	0.01	0.00	-11.10	0.02	0.18	0.01	0.45
Gateway 4 Series ^a	0.06	0.07	0.04	0.01	0.00	0.46	-11.99	0.18	0.01	0.41
HP Presario ^a	0.06	0.07	0.04	0.02	0.00	0.44	0.01	-12.08	0.01	0.39
IBM ThinkPad T ^a	0.04	0.07	0.05	0.02	0.00	0.35	0.01	0.16	-12.97	0.31
Toshiba Satellite ^a	0.07	0.07	0.03	0.01	0.00	0.51	0.02	0.18	0.01	-11.15

Notes: All elasticities computed in the last quarter of the sample, 2004Q2. The (i,j) cell reports the percentage change in the market share of product i (row) given a one percent increase in the price of product j (column). All products (desktops and laptops) have a Pentium 4 processor at the 2-2.99 GHz speed range. ^aIndicates a laptop.

(a) Table 7a: Estimated demand elasticities.

Willingness to pay				Markups and additional information	
Average consumer WTP (US\$)				Median markup (US\$)	76.4
1-1.49 GHz $\rightarrow$ 1.5-1.99 GHz				Median (p-mc)/p	0.078
1.5-1.99 GHz $\rightarrow$ 2-2.99 GHz				Corr(markup, price)	0.912
2-2.99 GHz $\rightarrow$ 3-3.99 GHz				Corr(ξ , ω)	0.820
Celeron $\rightarrow$ Pentium III					
Celeron $\rightarrow$ Pentium 4					
Celeron $\rightarrow$ Pentium M					
HP (Compaq) Presario					
Dell Inspiron					
Sony VAIO R					
IBM Thinkpad A					
1 year forward ^a					

Notes: ^aChange in willingness to pay over one year, see text.

(b) Table 7b: Markups and willingness to pay.

6.7 Figures 3–4: Heterogeneity in willingness to pay

Read: Figure 3 plots the distribution of willingness to pay for upgrading from Celeron to Pentium M. Figure 4 plots willingness to pay for moving one year forward with a fixed bundle.

Interpretation: Both figures show large heterogeneity. High-valuation consumers place substantial value on the frontier CPU upgrade, while the value of fixed characteristics erodes with time.

Economic message: Innovation has distributional effects: consumers differ sharply in how much they value frontier features and how quickly old bundles lose value.

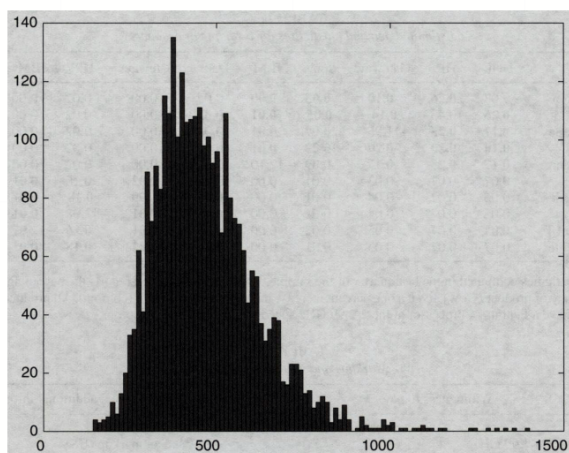


FIGURE 3
WTP for an upgrade from Intel's Celeron to its Pentium M chip (\$)

(a) Figure 3: WTP for Celeron-to-Pentium-M upgrade.

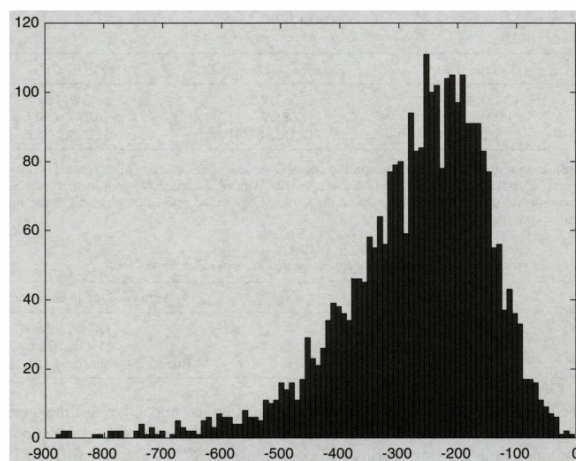


FIGURE 4
WTP for "1 year forward" (\$) (see text)

\$88. These estimates imply that per-configuration fixed costs consume a substanti

(b) Figure 4: WTP for one year forward.

6.8 Table 8: Bounds on fixed-cost parameters

Read: Table 8a reports fixed-cost intervals for Dell, HP, and Toshiba. Table 8b shows how estimates change under an alternative assumption that fixed-cost errors are uncorrelated with product choices.

Interpretation: Fixed costs are large enough to be economically important, and the estimation explicitly recognizes partial identification. The bounds are robust enough to support counterfactual analysis while avoiding false precision.

Economic message: Fixed costs explain why firms may eliminate older configurations even when some consumers still value them.

TABLE 8a
Bounds on fixed cost parameters F^d (\$ M)

	Estimated set	Estimated set (adjusted ^a)	95% CI ^b
F^{Dell}	[2.427, 4.529]	[2.353, 4.559]	[2.309, 4.914]
F^{HP}	[1.094, 2.721]	[1.081, 2.795]	[0.983, 3.009]
$F^{Toshiba}$	[2.622, 4.069]	[2.555, 4.119]	[2.295, 4.453]

Notes: This estimation utilizes information on specific notebook product lines of these firms (see text). The number of potential products, n^d , is 41 for Dell and Toshiba and 82 for HP. ^aWith adjustment for support bounds (see Section 4.2). ^bThe confidence interval does not take into account the variance due to the estimation of θ and to the simulation of expected variable profits, see text.

TABLE 8b
Set estimation of F^d assuming uncorrelated v errors

	Estimated set: baseline	Estimated set: uncorrelated errors
F^{Dell}	[2.427, 4.529]	[2.749, 4.213]
F^{HP}	[1.094, 2.721]	[1.495, 1.716]
$F^{Toshiba}$	[2.622, 4.069]	Bounds cross: lower=4.124, upper=3.509

Table 8: Bounds on fixed-cost parameters.

6.9 Tables 9–10: Effect of Intel’s Pentium M on outcomes and consumers

Read: Table 9 shows that removing Pentium M lowers notebook sales and raises desktop sales, implying substitution away from mobile PCs. Table 10 shows that Pentium M increases total consumer surplus from about 3.2% to 6.3%, with most of the gains accruing to the least price-sensitive consumers.

Interpretation: These tables are the main innovation-welfare results. Pentium M clearly increased aggregate welfare and notebook adoption, but its gains were highly uneven across consumer types.

Economic message: Upstream innovation can both expand market size and redistribute surplus toward consumers who value frontier quality most.

TABLE 9
Effect of Intel’s Pentium M on expected 2004Q2 outcomes

	Observed ^a	“No Pentium M” counterfactual	
		Lower bound ^b	Upper bound ^b
Total Notebook Sales (M)	1.707	1.379	1.614
Total Desktop Sales (M)	3.831	3.866	3.952
Mean Notebook price ^c (\$)	905	829	872
Impact on number of PC configurations (top 4 brands)			
	Observed	“No Pentium M” counterfactual	
		Lower bound ^b	Upper bound ^b
# P3_1.5-1.99	0	1	4
# C_1.5-1.99	3	0	4
# C_2-2.99	3	0	4
# P4_3-3.99	1	2	4
Impact on Pentium III’s share of total portables sales			
	Observed ^a	“No Pentium M” counterfactual	
		Lower bound ^b	Upper bound ^b
Share P3	0.077	0.156	0.239

^aAs explained in the text, these are not outcomes observed in the sample, but rather simulated expected outcomes computed at the set of products observed in the sample, which includes Pentium M-based notebooks. This does not pertain to the reported observed number of configurations, which is simply the observed sample quantity. ^bThe bounds represent the largest and smallest values computed over the set of all potential counterfactual equilibria. ^cSales-weighted average.

The effect of Intel’s Pentium M on consumers

	Observed ^a	“No Pentium M” counterfactual	
		Lower bound ^b	Upper bound ^b
Total expected consumer surplus	1213.7	1141.9	1176.3
Expected surplus for price-sensitivity quantiles (see text)			
0–20% price sensitive	992.7	932.6	955.1
20%–40% sensitive	130.3	123.1	129.9
40%–60% sensitive	76.5	73.2	76.8
60%–80% sensitive	12.6	11.9	12.8
80%–100% sensitive	1.6	1.1	1.7

Notes: All figures in M\$. ^aAs explained in the text, these are not outcomes observed in the sample, but rather simulated expected outcomes at the set of products observed in the sample. ^bThe bounds represent the largest and smallest values computed over the set of all potential counterfactual equilibria.

(a) Table 9: Effect of Pentium M on expected outcomes.

(b) Table 10: Effect of Pentium M on consumers.

6.10 Table 11: Expected welfare effects of adding older CPU products

Read: Table 11 evaluates adding back Pentium III- or Pentium 4-based notebooks to the 2004Q2 market. The consumer surplus and variable-profit gains are roughly offset by fixed costs, so total welfare cannot be clearly increased. But price-sensitive households gain because affordable notebooks become available.

Interpretation: The table shows the paper’s key distributional point. Keeping older products on the shelf does not necessarily raise total welfare, yet it allows the benefits of innovation to trickle down through lower prices and a wider variety of low-cost laptops.

Economic message: Product elimination can shut down a pro-consumer channel even when innovation raises aggregate surplus.

TABLE 11
Expected welfare effects of adding products based on older CPUs

	Added configurations	
	P3_1.5-1.99	P4_3-3.99
A. Total welfare components (M\$)		
Change to consumer surplus:	9.368	7.027
Change to PC maker variable profit:	2.802	1.705
Change to fixed costs:	[-14.267, -7.071]	[-10.149, -4.515]
Total effect:	[-2.097, 5.099]	[-1.417, 4.216]
B. Effect on different consumer segments' surplus (M\$)		
Quantile of price sensitivity		
0–20% price sensitive	6.018	6.445
20%–40% sensitive	2.060	0.443
40%–60% sensitive	0.978	0.118
60%–80% sensitive	0.213	0.015
80%–100% sensitive	0.099	0.005
C. Prices, sales, and choice probabilities in the Pentium III counterfactual		
Price range for added configurations (\$ US):	707–740	
Total sales (units) of added configurations:	141,988	
Increase in probability of buying a laptop, by consumer segment ^a :		
Quantile of price sensitivity		
0–20% price sensitive	27%	
20%–40% sensitive	18%	
40%–60% sensitive	24%	
60%–80% sensitive	27%	
80%–100% sensitive	6%	

Notes: Configurations based on Intel's Pentium III chips in the 1.5–1.99 GHz speed range were added to all four participating brands, while configurations based on its Pentium 4 chips with speed above 3 GHz were added to three (see text). All figures evaluated at expected 2004Q2 outcomes. ^aThese are not percentage point increases, but rather percentage increases in these probabilities.

Table 11: Welfare effects of adding products based on older CPUs.

6.11 Table A1: Robustness to dropping feasible-set criteria

Read: Table A1 examines how the feasible set and fixed-cost estimates change if the criteria for including CPU technologies are relaxed.

Interpretation: The estimates change somewhat, but the basic fixed-cost bounds and qualitative conclusions are robust to the inclusion criteria.

Economic message: The main counterfactual conclusions are not driven by one arbitrary rule defining feasible CPU technologies.

7 Interpretive synthesis

- **Innovation has two margins.** The direct margin is better frontier products; the indirect margin is the disappearance of older products.

TABLE A1

Robustness to dropping criteria for inclusion in the feasible set

A. Technologies added to the set when dropping each criterion ^a				
Quarter	(i)	(ii)	(iii)	
Q7	PM 1.0-1.49	PM 1.5-1.99		
Q8	P4 1.0-1.49, PM 1.0-1.49			
Q9	P4 1.0-1.49			
Q10	P4 1.0-1.49			
Q11	P3 1.0-1.49		P3 1.5-1.99	
Q12	P3 1.0-1.49		P3 1.5-1.99	
B. Estimated sets obtained by dropping inclusion criteria ^b				
Parameter	Dropped criterion			
	None	(i)	(ii)	(iii)
F^{Dell}	[2.427, 4.529]	[2.449, 4.909]	[2.132, 4.443]	[2.482, 4.726]
F^{HP}	[1.094, 2.721]	[1.061, 2.779]	[1.094, 2.636]	[1.079, 2.586]
$F^{Toshiba}$	[2.622, 4.069]	[2.558, 3.999]	[2.417, 4.040]	[2.575, 4.187]

^aNotes: The first, second and third columns show which technologies are added to the feasible set for each criterion.

^bNotes: The first, second and third columns show the estimated sets obtained by dropping each criterion.

Table A1: Robustness to feasible-set criteria.

- **Variety matters for distribution.** The consumers who benefit most from frontier technology are not necessarily the consumers who would benefit most from preserving older low-price varieties.
- **Fixed costs rationalize product elimination.** Downstream firms do not keep every positive-variable-profit product because each configuration requires fixed costs.
- **Multiple equilibria matter.** Product-choice games need not have unique equilibria, so counterfactuals should acknowledge potential-equilibrium sets rather than pretending point predictions are available.
- **Policy is delicate.** Forcing old products to remain available can help price-sensitive consumers, but the aggregate welfare case is ambiguous because fixed costs absorb much of the surplus gain.

8 Short summary

- The paper studies the welfare effects of CPU innovation in the U.S. home PC market.
- It estimates a model in which PC makers endogenously choose product portfolios and then set prices.
- Demand estimates show strong heterogeneity in price sensitivity and taste for portability and CPU technology.
- Fixed costs are economically important and partially identified because product choices depend on unobserved fixed-cost shocks.
- The Pentium M increased notebook sales and aggregate consumer surplus.
- The benefits of Pentium M were concentrated among the 20% least price-sensitive consumers.
- Pentium M crowded out older Pentium III and Pentium 4 notebook configurations.
- Adding older configurations back to the market does not clearly raise total welfare but helps price-sensitive households by creating affordable laptops.
- The incremental contribution is to connect upstream innovation, endogenous downstream product variety, partial identification, and distributional welfare.

9 Conclusion

9.1 Core conclusion

Eizenberg shows that upstream innovation can raise aggregate welfare while simultaneously narrowing downstream product variety. In the U.S. home PC market, Intel’s Pentium M improved mobile computing and increased consumer surplus, but it also crowded out older notebook configurations that would have served price-sensitive households.

9.2 Economic intuition

Innovation changes the feasible set for downstream firms. Because offering each configuration involves fixed costs, firms do not keep every older product available. When they drop basic configurations, the benefits of frontier innovation fail to reach consumers with low willingness to pay for frontier features. Product variety therefore mediates the distributional impact of technological progress.

9.3 Incremental contribution in one sentence

The paper’s distinctive contribution is to show, using a structural model with endogenous product portfolios and partially identified fixed costs, that upstream innovation can have large aggregate benefits while also shutting down affordable downstream varieties that would spread those benefits to price-sensitive consumers.

9.4 Final takeaway

Innovation policy and competition policy should not evaluate new technologies only by aggregate surplus. They should also ask how innovation changes the product menu and which consumer groups gain access to the new frontier or to discounted older technologies.